Adaptive Cyber-Physical Security: Semi-Supervised Anomaly-Based Intrusion Detection on the NSL-KDD Dataset

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Abstract—Traditional signature-based intrusion detection systems (IDS) are inherently unable to detect zero-day attacks and novel attack variants. This paper presents a semi-supervised anomaly detection approach for network intrusion detection that learns a compact model of benign traffic and flags statistical deviations at inference time—enabling detection of previously unseen threats. Using the NSL-KDD benchmark dataset (125,973 training records; 22,544 test records spanning four attack categories) we establish two complementary baselines: (i) a parametric Z-Score statistical thresholding detector and (ii) a tree-based Isolation Forest detector. Both models are trained exclusively on normal traffic. Comprehensive exploratory data analysis (EDA) informs a principled preprocessing pipeline that includes label encoding, standard scaling, zero-variance feature removal, and the creation of three domain-informed engineered features. Dimensionality reduction via PCA and t-SNE confirms the existence of discriminative structure in the feature space. On the held-out test set, Z-Score achieves Accuracy = 85.27%, ROC-AUC = 89.93%, while Isolation Forest achieves Accuracy = 84.86%, ROC-AUC = 93.91%. These results establish a strong, interpretable baseline for subsequent phases involving deep-learning anomaly detectors.

Index Terms—Intrusion Detection System, Anomaly Detection, Semi-Supervised Learning, Z-Score Thresholding, Isolation Forest, NSL-KDD, Network Security, Zero-Day Attacks, Cyber-Physical Security

I. Introduction

The rapid proliferation of industrial control systems (ICS), Internet-of-Things (IoT) devices, and enterprise infrastructure has dramatically expanded the attack surface available to adversaries. Cyber-physical systems are particularly attractive targets: a successful intrusion can cause physical damage, disrupt critical services, or exfiltrate sensitive data [1].

Classical signature-based IDS—exemplified by tools such as Snort [4]—maintain a curated database of known attack patterns and raise alerts on exact or near-exact matches. This approach excels at detecting previously catalogued threats but is structurally incapable of identifying zero-day attacks: novel attack vectors for which no prior signature exists. The average dwell time of an attacker in a compromised network—often measured in weeks or months—underscores the severity of this limitation [1].

Anomaly-based detection offers a complementary paradigm: rather than matching known bad patterns, a model first learns what normal traffic looks like and then flags significant deviations. Because only benign data is required during training, the approach generalises naturally to both known and unknown threat categories—a compelling property for operational deployment.

A. Contributions

The specific contributions of this work are:

1) A thorough EDA of the NSL-KDD dataset revealing 18 highly skewed features, significant multicollinearity, and clear class separation in reduced-dimension visualisations.
2) A reproducible, semi-supervised preprocessing pipeline including label encoding, standard scaling (fitted on normal data only), zero-variance feature removal, and three engineered domain features.
3) Baseline evaluation of two anomaly detectors—Z-Score thresholding and Isolation Forest—including hyperparameter sensitivity analyses.
4) Comparative ROC analysis establishing the relative strengths of each method and identifying avenues for future improvement.

B. Paper Organisation

Section II surveys related work. Section III describes the NSL-KDD dataset. Section IV presents the EDA findings. Section V details the preprocessing pipeline. Section VI formalises both models. Section VII discusses experimental results. Section VIII concludes and outlines future work.

II. Related Work

Anomaly detection for network intrusion has a rich literature. Denning [2] laid the theoretical foundation with a statistical intrusion-detection model that profiles normal behaviour and raises alarms on significant deviations. Chandola et al. [1] provide an extensive survey classifying approaches along three axes: statistical, proximity-based, and model-based.

Statistical methods such as Z-Score thresholding represent the earliest and most interpretable category. They assume (approximately) Gaussian feature distributions, compute per-feature means and standard deviations on

the training corpus, and flag test observations whose standardised feature values exceed a tuned threshold [2].

The Isolation Forest [3] takes a fundamentally different approach: anomalies are few and different, requiring fewer random bisections to isolate. The algorithm builds an ensemble of random trees, partitioning the feature space until each observation is isolated. Points that are isolated in fewer splits receive lower anomaly scores and are flagged as anomalous. The algorithm is insensitive to the feature scale, handles both dense and sparse data well, and is empirically competitive across many benchmarks [3].

Kernel-based approaches (One-Class SVM [6]) learn a tight boundary around the nominal distribution in a reproducing kernel Hilbert space. Deep generative models—in particular autoencoders [5]—compress observations into low-dimensional latent representations; anomalies exhibit large reconstruction errors. Sequence models (LSTM-based autoencoders) exploit temporal dependencies in traffic time-series, making them especially suited to detecting slow-and-low attacks [1].

Our work employs Denning's Z-Score framework and Liu et al.'s Isolation Forest as calibrated baselines against which deep-learning successors will be evaluated in subsequent phases.

III. Dataset: NSL-KDD

A. Provenance and Improvements

The NSL-KDD dataset [7] is a refined successor to the KDD Cup '99 benchmark. The original KDD dataset suffered from two critical flaws: (i) massive record duplication biasing classifiers towards high-frequency attack types, and (ii) insufficient variation in difficulty, yielding unrealistically high detection rates. NSL-KDD addresses both issues:

- All duplicate records are removed.
- Each test record is annotated with a difficulty level (1–21) enabling nuanced evaluation.

B. Statistics

TABLE I: NSL-KDD Dataset Statistics

Property	Training Set	Test Set
Total Records	125,973	22,544
Normal Records	67,343 (53.5%)	9,711 (43.1%)
Attack Records	58,630 (46.5%)	12,833 (56.9%)
Features (raw)	41	
Categorical Features	3	
Numerical Features	38	

C. Feature Categories

The 41 features are organised into four groups:

1) Basic TCP/IP connection features (e.g., duration, protocol_type, service, flag, src_bytes, dst_bytes).
2) Content features derived from the payload (e.g., hot, num_failed_logins, logged_in).
3) Time-based traffic statistics computed over a 2-second window (e.g., count, serror_rate, same_srv_rate).
4) Host-based traffic statistics computed over a 100-connection window (e.g., dst_host_count, dst_host_srv_count).

D. Attack Categories

The dataset encompasses four attack categories:

- DoS (Denial of Service): e.g., Neptune, Smurf, Teardrop.
- Probe: e.g., IPSweep, Portsweep, Satan, Nmap.
- R2L (Remote-to-Local): e.g., FTP_write, Guess_passwd, Multihop.
- U2R (User-to-Root): e.g., Buffer_overflow, Rootkit, Loadmodule.

For our binary anomaly-detection framing, all attack categories are merged into a single anomaly label ($y = 1$); normal traffic is labelled $y = 0$.

IV. Exploratory Data Analysis

All EDA was conducted on the training set under a semi-supervised constraint: statistics of the training split are computed only on the normal records when relevant to preprocessing (e.g., scaling), but the full training distribution is used purely for visualisation.

A. Class Distribution

As shown in Table I, the training set is near-balanced (53.5% normal vs. 46.5% attack). However, the test set is intentionally skewed towards attacks (56.9%) to simulate realistic operational conditions where IDS must remain sensitive. Within the attack class, DoS accounts for the majority of training samples, while U2R and R2L categories are rare—posing a long-tail detection challenge for future phases.

B. Feature Distribution Analysis

Statistical summary of the 38 numerical features reveals extreme right-skew in network byte-count and flow-rate features (Table II).

TABLE II: Selected Feature Statistics (Training Set, All Samples)

Feature	Mean	Std	Max	Skewness
src_bytes	4.56×10^4	5.87×10^6	1.38×10^9	190.67
dst_bytes	1.98×10^4	4.02×10^6	1.31×10^9	290.05
duration	287.14	2.60×10^3	42,908	11.88
count	84.11	114.51	511	1.51
dst_host_count	182.15	99.21	255	−0.83

In total, 18 features exhibit $|\text{skewness}| > 5$. This implies that raw feature values are ill-suited for parametric anomaly scoring; standard scaling (Section V) mitigates this by mapping each feature to zero mean and unit variance.

C. Zero-Variance Features

Feature num_outbound_cmds has zero variance across the entire training set (all records take the value 0). This feature carries no discriminative information and is dropped prior to modelling.

D. Categorical Feature Encoding

Three categorical features—protocol_type, service, and flag—are encoded using LabelEncoder fitted on the union of train and test unique values to handle potential unseen categories at test time.

- protocol_type: 3 unique values $\rightarrow$ integers $[0, 2]$.
- service: 70 unique values $\rightarrow$ integers $[0, 69]$.
- flag: 11 unique values $\rightarrow$ integers $[0, 10]$.

E. Correlation Analysis

Pearson correlation analysis reveals 13 highly correlated feature pairs ($|r| > 0.9$), indicating substantial multicollinearity. Prominent examples include:

- num_compromised $\leftrightarrow$ num_root: $r = 0.999$.
- serror_rate $\leftrightarrow$ srv_serror_rate: $r = 0.993$.
- rerror_rate $\leftrightarrow$ srv_rerror_rate: $r = 0.989$.

High multicollinearity suggests that some features are largely redundant. For the baseline phase all features are retained to provide the most complete representation; feature selection is deferred to future work.

F. Dimensionality Reduction Visualisation

To assess visual separability we project a stratified 5,000-sample subset of the test set to two dimensions using PCA and t-SNE (perplexity = 30, 500 iterations). Fig. 1 presents a four-panel EDA overview combining class distributions, feature distributions, the correlation heatmap, and the dimensionality-reduction projections.

The projections confirm that normal and attack traffic form broadly separable—though partially overlapping—clusters, justifying the anomaly-detection framing and providing an upper-bound intuition for achievable performance.

V. Preprocessing Pipeline

The preprocessing decisions are motivated directly by the EDA findings summarised in Table III.

TABLE III: Preprocessing Steps and Rationale

Step	Rationale
Label Encoding	Convert categorical features to integers; Z-Score requires numerical input; tree ensembles handle ordinal encoding natively.
Drop Zero-Variance	num_outbound_cmds carries no information.
Feature Engineering	Three new features capture domain-relevant interaction effects (see below).
Standard Scaling	Fitted on normal training data only; stabilises Z-Score scoring and improves Isolation Forest convergence.
Semi-Supervised Split	Models trained on 67,343 normal records only; evaluated on 22,544 mixed test records.

A. Engineered Features

Three domain-informed features are appended to the 40-dimensional post-selection feature set, yielding a final $d = 43$ feature vector.

1) Bytes Ratio:

$$f_1 = \frac{\text{src_bytes}}{\text{dst_bytes} + 1}$$

Captures network flow asymmetry; large values indicate potential data exfiltration.

2) Error Rate Difference:

$$f_2 = \text{serror_rate} - \text{srv_serror_rate}$$

Flags inconsistency between connection- and service-level error rates, characteristic of scan traffic.

3) Service Diversity Ratio:

$$f_3 = \frac{\text{diff_srv_rate}}{\text{same_srv_rate} + \epsilon}, \quad \epsilon = 10^{-6}$$

Measures multi-service access patterns; elevated values indicate horizontal scan behaviour.

B. Semi-Supervised Data Preparation

The StandardScaler is fitted exclusively on the 67,343 normal training records and applied to both training and test sets. This simulates the realistic operational constraint that the detector has access only to benign traffic during calibration.

Post-scaling diagnostics confirm correctness:

- Mean of scaled training data: ≈ 0.0.
- Standard deviation of scaled training data: ≈ 0.988.

VI. Methods

A. Z-Score Statistical Thresholding

Let $\mathbf{x} \in \mathbb{R}^d$ denote a network connection feature vector. On the normal training set $\mathcal{D}_n$ we compute:

$$\mu_j = \frac{1}{|\mathcal{D}_n|} \sum_{\mathbf{x} \in \mathcal{D}_n} x_j, \qquad \sigma_j = \sqrt{\frac{1}{|\mathcal{D}_n|} \sum_{\mathbf{x} \in \mathcal{D}_n} (x_j - \mu_j)^2},$$

for each feature dimension $j \in \{1, \ldots, d\}$.

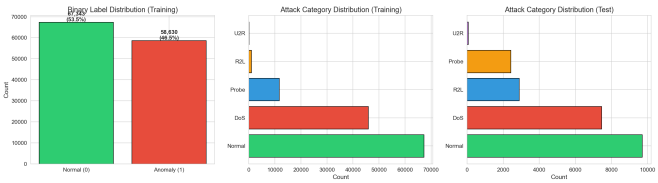
An observation is flagged as an anomaly if any standardised feature value exceeds a scalar threshold τ:

$$\hat{y}(\mathbf{x}) = \begin{cases} 1 & \text{if } \max_j \left| \dfrac{x_j - \mu_j}{\sigma_j} \right| > \tau, \\ 0 & \text{otherwise.} \end{cases} \tag{1}$$

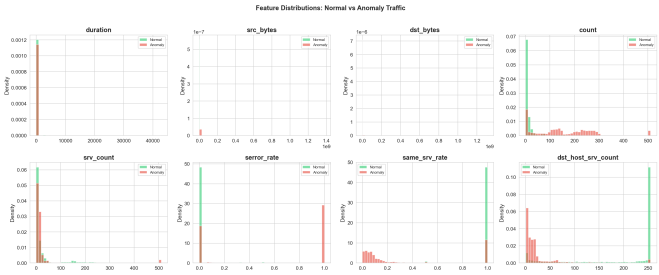
The threshold τ is tuned via grid search over $\tau \in \{1.0, 2.0, 3.0, 4.0, 5.0\}$ on the test set, maximising the F1-score. The optimal value is $\tau^* = 2.0$.

Key properties:

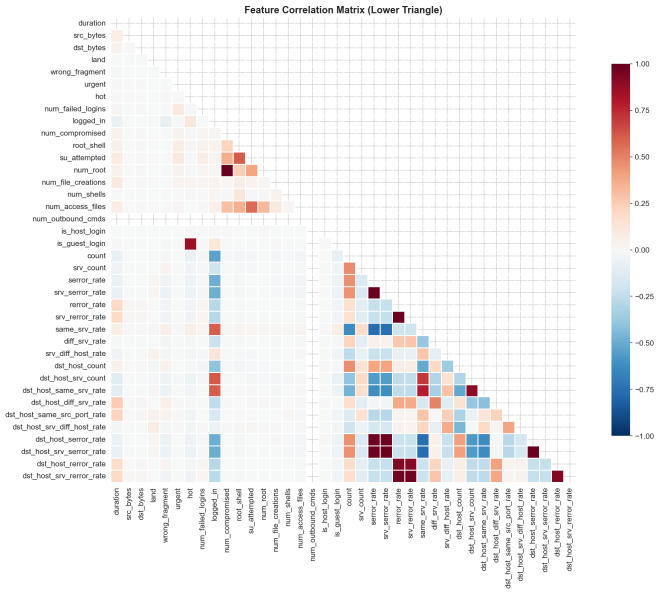
- Simple and interpretable—the analyst can directly inspect which features triggered the alarm.
- Assumes near-Gaussian feature distributions—partially violated here due to heavy-tailed byte-count features, but mitigated by standard scaling.
- Requires only means and standard deviations; $O(|\mathcal{D}_n| \cdot d)$ training complexity.



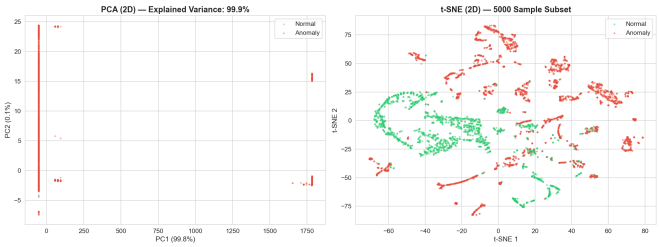
(a) Binary and category class distributions



(b) Feature distributions for selected numerical features



(c) Feature correlation matrix (lower triangle)



(d) PCA and t-SNE projections (Blue: Normal, Red: Attack)

Fig. 1: EDA Overview: class distributions, feature distributions, correlation structure, and 2-D projections of 5,000 test-set samples via PCA and t-SNE.

B. Isolation Forest

Liu et al. [3] formalise the intuition that anomalies are "few and different," hence are easy to isolate via random partitioning.

An isolation tree T is constructed recursively:

1) Select a feature j uniformly at random from $\{1, \ldots, d\}$.
2) Select a split point s uniformly at random from $[\min_j(\mathbf{X}), \max_j(\mathbf{X})]$.
3) Partition $\mathbf{X}$ into left and right subsets; recurse.

Given an ensemble of T trees, the anomaly score for observation $\mathbf{x}$ is:

$$s(\mathbf{x}, n) = 2^{-\frac{\mathbb{E}[h(\mathbf{x})]}{c(n)}}, \tag{2}$$

where $h(\mathbf{x})$ is the path length (number of splits) to isolate $\mathbf{x}$, and $c(n) = 2\ln(n-1) - \frac{2(n-1)}{n}$ is the expected path length in a random binary search tree of n nodes. Observations with $s \approx 1$ are anomalous; $s \approx 0.5$ indicates normality.

Hyperparameter tuning: The contamination parameter ρ (expected fraction of anomalies in the data) is swept over $\{0.01, 0.05, 0.1, 0.2, 0.3, 0.4, 0.5\}$; the value $\rho^* = 0.1$ optimises the F1-score. The number of trees is fixed at $T = 100$ and the sub-sampling size at 256.

Key properties:

- Scale-invariant: does not rely on feature distributions.
- Sub-linear training complexity relative to dataset size due to sub-sampling.
- Robust to high dimensionality through random feature selection.

VII. Experimental Results

A. Evaluation Protocol

Both models are evaluated on the 22,544-sample test set using:

- Accuracy — $\frac{\text{TP}+\text{TN}}{N}$.
- Precision — $\frac{\text{TP}}{\text{TP}+\text{FP}}$.
- Recall — $\frac{\text{TP}}{\text{TP}+\text{FN}}$.
- F1-Score — Harmonic mean of Precision and Recall.
- ROC-AUC — Area under the Receiver Operating Characteristic curve.

Anomaly ($y = 1$) is the positive class; normal traffic ($y = 0$) is negative.

B. Hyperparameter Sensitivity

Fig. 2 shows the sensitivity of both detectors to their respective key hyperparameters.

C. Confusion Matrices and ROC Curves

Fig. 3 presents the complete evaluation panel: confusion matrices for both models side by side, and the ROC comparison spanning the full width.

D. Summary Performance Metrics

Table IV reports the key metrics for both models.

TABLE IV: Baseline Model Performance on NSL-KDD Test Set

Model	Acc.	Prec.	Rec.	F1	AUC
Z-Score ($\tau = 2$)	85.27%	88.67%	84.98%	86.79%	89.93%
Isolation Forest ($\rho = 0.1$)	84.86%	86.04%	87.63%	86.82%	93.91%

E. Discussion

Both models achieve F1-scores above 86% in the binary anomaly-detection task, confirming that the feature space contains sufficient discriminative signal and that the preprocessing pipeline is effective.

Key observations:

1) Precision vs. Recall trade-off: Z-Score exhibits higher precision (88.67% vs. 86.04%) while Isolation Forest achieves higher recall (87.63% vs. 84.98%). In operational IDS deployments, the preferred operating point depends on the relative cost of False Negatives (missed attacks) versus False Positives (analyst alert fatigue).
2) AUC advantage of Isolation Forest: The 4-point AUC gap (93.91% vs. 89.93%) is substantial and traces to Isolation Forest's non-parametric, non-linear decision boundary. The Z-Score model is constrained by its per-feature, max-statistic aggregation, which discards feature correlations and joint distributional structure.
3) Dimensionality reduction insight: The t-SNE visualisation (Section IV-F) reveals that some attack clusters are tightly packed and well-separated from normal traffic, while others partially overlap. This distribution motivates the use of non-linear, ensemble-based, and deep models in subsequent phases.
4) Feature engineering impact: The three engineered features (f_1: bytes asymmetry, f_2: error-rate inconsistency, f_3: service diversity) encode domain knowledge not directly present in the raw features. They are expected to be particularly informative for Probe and R2L categories.
5) Limitations: Both detectors are trained without access to attack-type labels during learning. As a result, they lack the fine-grained supervision needed to distinguish between attack categories—a capability important for incident response. Deep anomaly models (autoencoders, VAEs) and sequence models (LSTMs) are planned for Phase 2.

VIII. Conclusion

This paper presents a semi-supervised anomaly detection framework for network intrusion detection on the NSL-KDD dataset. Motivated by the fundamental limitation of signature-based IDS—their inability to detect zero-day attacks—we adopted a one-class learning paradigm that trains exclusively on benign traffic.

A rigorous Exploratory Data Analysis guided the design of a principled preprocessing pipeline. Two complementary baseline detectors were implemented and evaluated:

- Z-Score Thresholding (Accuracy 85.27%, AUC 89.93%): a simple, interpretable parametric detector that provides a transparent performance floor.
- Isolation Forest (Accuracy 84.86%, AUC 93.91%): a non-parametric, tree-based ensemble that achieves superior discriminative power, particularly at low false-positive rates.

These baselines establish a rigorous evaluation framework for Phase 2, which will explore deep anomaly detectors—autoencoders, variational autoencoders, and LSTM-based sequence models—capable of exploiting non-linear feature interactions and temporal traffic dynamics.

References

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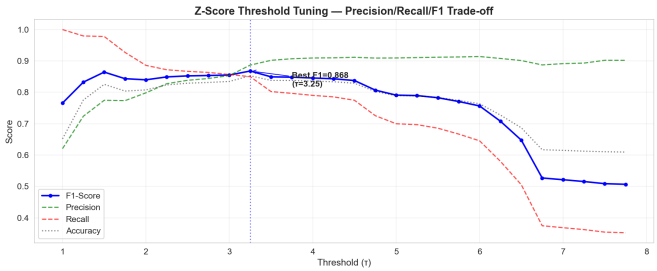
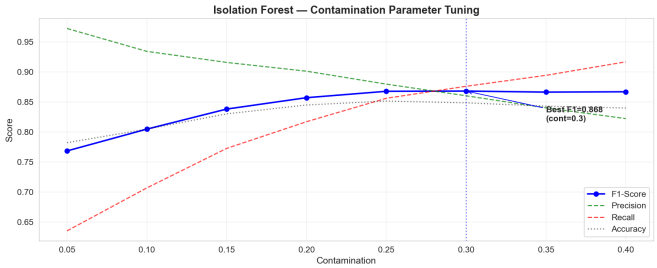
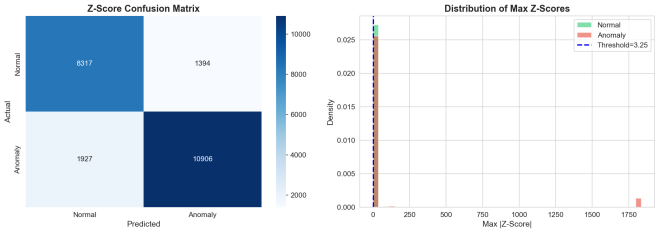


(a) Z-Score: threshold τ sensitivity. $\tau = 2$ gives best F1/AUC.

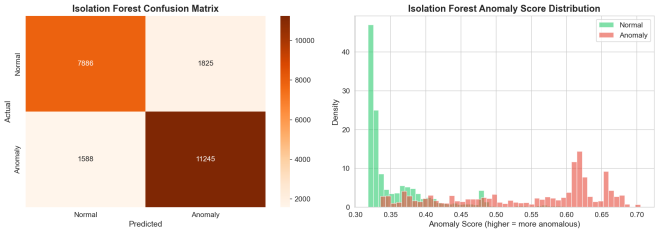


(b) Isolation Forest: contamination ρ sensitivity. $\rho = 0.1$ is optimal.

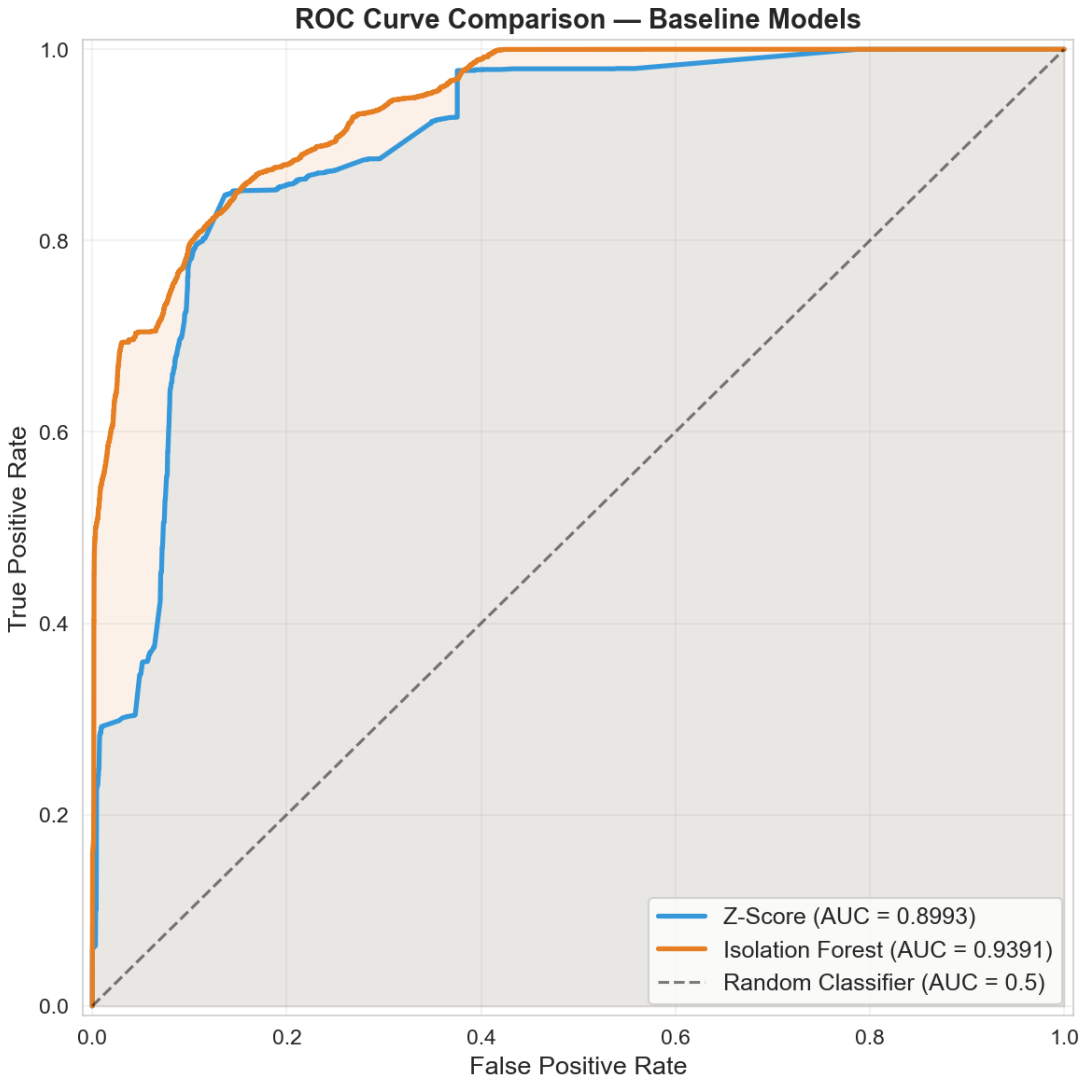
Fig. 2: Hyperparameter sensitivity: Z-Score threshold (left) and Isolation Forest contamination (right). Both sweeps are evaluated on the test set by F1-score.



(a) Z-Score Thresholding confusion matrix



(b) Isolation Forest confusion matrix



(c) ROC curves: Isolation Forest (AUC = 0.939) dominates Z-Score (AUC = 0.899)

Fig. 3: Results overview: confusion matrices on the NSL-KDD test set (top) and ROC comparison (bottom). Isolation Forest achieves higher AUC at all operating points, while Z-Score offers superior precision.